Research Plan

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Introduction

Literature Review

Although many different types of gas detectors are available for purchase, few of them will be able to detect different types of the main types of gases commonly found in gas leaks. These detectors will also tend to be expensive and some can't work in the case of a power outage.

These harmful gases will include propane, butane, LPG (a mixture of propane and butane), and carbon monoxide. While carbon monoxide leaks may be rare, they will still pose serious threats. For instance, in April 2020, an engaged couple in South Africa tragically passed away due to carbon monoxide poisoning at the Kliphuis Guesthouse near Kareedouw in the Eastern Cape. They will have been discovered unresponsive in the shower by a friend who joined them on their trip.

To develop the gas detector, a wide range of electronic components will be required, along with skills in program development and general coding—particularly using the programming language C++. Initial research will identify the necessary components and programming tools; however, additional investigation will be needed to determine how all components will be connected to an Arduino-compatible microcontroller, and how full system compatibility will be achieved.

In preparation for the research project, information will be gathered from various websites and Artificial Intelligence tools. Questions regarding component compatibility and part selection will be addressed through this research. If necessary, a professional in the field will be consulted for further guidance.

The physical and technical research will stand out by focusing on a real-world safety issue rather than simply proving functionality. It will reflect a deliberate effort to evaluate and select each component for optimal performance and compatibility in the final product.

Key Terms

- Arduino – An open-source electronics platform based on user-friendly hardware and software; essentially, a microcontroller board.
- V - Voltage
- ESP32 – A low-cost, low-power system-on-a-chip (SoC) with integrated Wi-Fi and Bluetooth, commonly used in Internet of Things (IoT) projects.
- Python – A high-level, general-purpose programming language known for its readability and simplicity.
- LCD – A flat-panel display that uses liquid crystals to control light and display information.
- 12V DC Adapter – Converts high-voltage AC (from a wall outlet) into 12V direct current (DC).
- 12V Li-ion or LiPo Battery – Rechargeable battery packs providing approximately 12 volts of DC power.
- LiPo Charger 3.7V with DC-DC Boost 4.5–24V – A module that charges single-cell 3.7V LiPo batteries and boosts output voltage.
- Voltage Regulator (Buck Converter) – A step-down converter that regulates voltage to match component requirements. \
- DC Barrel Jack (2.1mm / 5.5mm) – A common power connector used for 12V DC input.
- JST Connectors – Compact, reliable wire-to-board or wire-to-wire connectors.
- DC Jack to Alligator Clips – An adapter cable that converts a DC jack into clip terminals for flexible power connections.
- Standard Electrical Wires (AWG Ratings) – Wire gauges that determine wire thickness and current-carrying capacity.
- LPG - Liquefied Petroleum Gas, is a flammable hydrocarbon gas mixture primarily composed of propane (C_3H_8) and butane (C_4H_{10})
- VCC -Voltage at the Common Collector, and it represents the positive power supply voltage used in circuits
- Ground - Acts as the return path. Makes the connection to negative terminal.
- Voltage Divider - When a voltage is too high for a component you can seperate it via using two resistors in series. These resistors will have two seperate volatges over them, these volatges add up to the total voltage. Thus the voltage was divided between resistors.

- Analog Output/Pin - Each gas sensor has this pin. It sends a voltage out based on how much of a specific gas it detects in the air. If it is designed to detect specifically CO like the MQ 7 gas sensor, it will be more sensitive to this gas in the air and send a higher voltage if there is a high amount of CO.

This research will provide a foundation for future upgrades, support the construction phase, and enable efficient technical planning. Using the knowledge and skills gained, the team will strive to develop a gas detector that will reliably identify gas leaks in both wet and dry environments—even during power outages.

Need / Problem Statement

LPG, and other gasses (such as propane or butane) will be the primary fuels used in gas stoves globally. Carbon monoxide will be formed when these carbon-based fuels burn incompletely—often due to poor ventilation or faulty combustion.

In their pure forms, these gases will be odorless, colorless, and invisible—making them undetectable without the help of sensing technology. Although odorants will be added to assist detection, they may fail in critical situations—such as when someone accidentally leaves the stove on, walks away while cooking, or has congenital anosmia (a condition that results in an inability to smell).

The target users for this project will include:

- Individuals with a reduced or absent sense of smell.
- People who rely on gas stoves or gas-powered appliances.
- Communities affected by frequent power outages or limited access to electricity.
- South Africans who experience loadshedding.

Without a method to actively monitor gas leaks, the consequences could range from mild health effects to severe injury or even death.

Research Questions

What microcontroller is the most effective for this specific project?

How to convert the APG 12V battery to 5V for the microcontroller or 3.3V

How to manage the different voltages required for different components.

What language or IDE would be best to use?

Will the sensors be able to detect minimal amounts of CO.

Aim

This project will aim to develop a simple, affordable, and effective gas detector capable of identifying leaks from LPG, carbon monoxide, and other gasses. It will also be able to operate without electricity for use in communities where power outages are frequent and where gas stoves are still commonly used. It will be able to detect LPG from gas stoves in households.

Engineering Goals

The primary engineering goal will be to reduce injuries and fatalities caused by gas leaks by components such as gas stoves. This will be achieved by offering a cost-effective, multi-gas detection system, in contrast to expensive commercial detectors that only identify a few gasses.

The gas detector must also be able to run without electricity via a built in battery within the gas detector supplying it with power when there is a outage, relevant to South Africa's loadshedding. This is especially important in communities where power outages are common.

The buzzer alarm must also be loud enough to be heard from across a house to ensure that the residents will definitely know when there is a gas leak.

Method

Materials

Final List Of Materials:

- Battery LiPo 1000 mAh 7.4V
  - https://www.robotics.org.za/603450?search=lipo%20battery
- MQ 5 Toxic Gas Sensor:
  - https://www.winsen-sensor.com/d/files/MQ-5.pdf
- Robotico ESP32 Development Board 2.4GHz Dual-Core WiFi + Bluetooth
  - https://www.takealot.com/robotico-esp32-development-board-2-4ghz-dual-core-wifi-bluetooth/PLID72203100
  - Cheap microcontroller with a big variety of features. Will act as the core/controller for the gas detector.
- Buck Converter [BMT ADJ DC/DC MODULE 3A+DISPLAY]:
  - https://www.communica.co.za/products/bmt-adj-dc-dc-module-3a-display
  - Can input a larger voltage and output a smaller adjustable voltage. This is used to step down the 12V from the battery to 5V for the microcontroller.
- MF 1/4W 1% E12 RES KIT
  - https://www.communica.co.za/products/mf-1-4w-1-e12-res-kit?variant=31251775357001
  - A resistor kit which contains the required resistors to make a voltage divider.
- Multipurpose 12V DC 2AMP Power Supply - AC 240V
  - https://www.takealot.com/multipurpose-12v-dc-2amp-power-supply-ac-240v/PLID46906813?gad_campaignid=20213916716&gad_source=1&gclid=Cj0KCQjw5JXFBhCrARIsAL1ckPu2BIxR7SLa2u_vU9od8uVXzJ7s0YKz-su71M2onFX_yjZEbW4BEp0aAk3QEALw_wcB&gclsrc=aw.ds
- DC Barrel Power Jack, Female, SMD, 2155 Standard, 4 Pack
  - https://www.robotics.org.za/connectors/dc-jack-connectors/DC050-2155-SMD

- HKD BREADBOARD 8,3X5,5CM 400TP
  - https://www.communica.co.za/products/hkd-breadboard-8-3x5-5cm-400tp?variant=31932627419209
  - Allows for more convenient connection of components.
- USB CABLE 1,5M AM-MICRO
  - https://www.communica.co.za/products/usb-cable-1-5m-am-micro?variant=17185748779081
  - Used to input 5V to the microcontroller via its usb 2.0 compatible port.
- HKD ELECTRONIC BUZZER
  - https://www.communica.co.za/products/hkd-electronic-buzzer?variant=47729630839084
  - Buzzer that can work with both 3.3V and 5V.
- HKD RIBBON JUMPER 40W M/M 30CM
  - https://www.communica.co.za/products/hkd-ribbon-jumper-40w-m-m-30cm?variant=31916109561929
  - Wires used to make connections.
- HKD BREADBOARD JUMPER KIT (140)
  - https://www.communica.co.za/products/hkd-breadboard-jumper-kit-140?variant=31932617097289
  - More wires to connect components.
- A Soldering Set
  - Used to make connections more secure via soldering the pins of components to wires.
  - Already have a soldering set.
  - https://www.takealot.com/16-piece-soldering-kit-with-meter-and-stand-adjustable-heat-200-/PLID91983733?srsltid=AfmBOopdpGcytro52_7AHngCR2IIXgmtrVREhXElgu_z7g_c_tnUN4bMB9U
- Plastic Tubing used to make the wiring cleaner and more clear to see what its purpose is. Black is used to show it is for ground, Red for VCC and Blue for control or input and output.
  - Supplied by Riaan Snyman
- ABS Enclosure 115 x 75 X 52 with Mounting Tags, Grey
  - https://www.robotics.org.za/A3-TAG-GREY?search=abs%20enclosure%20115
- Universal 2A LiPo Charger, 4.2V / 7.4
  - https://www.robotics.org.za/TP5100?search=Lithium%20Polymer%20charger

Previous old list:

- Gas Sensors (MQ-9) – To detect carbon monoxide and flammable gases.
- Microcontroller (ESP32) – Supports C++ and provides processing power.
- LCD Display (2.8" Capacitive Touch) – For user interface and gas level readouts.
- Buzzer or LED Indicator – To trigger visual or audible alerts in unsafe conditions.
- Water-Resistant Enclosure – To protect components from humidity or splashes.
- Power Supply (12V DC Adapter) – For stable voltage from mains electricity.
- Rechargeable Battery (12V Li-ion or LiPo) – Backup power during outages.
- Battery Management Module (3.7V LiPo Charger with DC-DC Boost) – For regulated charging and power delivery.
- Voltage Regulator (Buck Converter) – To reduce and stabilize voltage output.
- DC Barrel Jack (2.1mm / 5.5mm) – For connecting standard adapters to the power circuit.
- JST Connectors – For compact and reliable connections between modules.
- Battery Connector Kit (50A or 120A) – For quick-connect backup battery integration.
- DC Jack to Alligator Clips – For flexible testing with variable power sources.
- Standard Electrical Wires (18–22 AWG and 14–16 AWG) – To distribute power effectively depending on current demand
- ENX4LBS Enertec Motorcycle battery
- MXS 10 Charger
- MQ 7 Toxic Gas Sensor

Procedure

To meet its functional requirements, the gas detector will:

- Detect multiple target gases: Carbon Monoxide, Propane, Butane, LPG, and Methane.
- Be cost-effective and accessible to users across different income levels.
- Function during power failures, thanks to an integrated rechargeable battery system.
- Be able to turn the buzzer on when there is too much of a harmful gas that it detects.
- Be able to run without power for 6 hours.

The Gas Detector will be able to be powered via house power. The charger will charge the battery from the house power with 12V. Then the buck converter will change the voltage to 5V which is what the Gas Detector will work with.

An estimated time of 9 days will be allocated for the full construction and development process:

- 7 days for building the gas detector as well as for programming the gas detector.
- 2 days for the testing of the gas detector.

Previous Procedure That Was Changed:

The detector will receive primary power through a 12V wall adapter, with voltage regulation circuitry tailoring output for each component's needs.

The majority of parts will be sourced from robotics.org.za, but some items may need to be acquired locally, depending on availability.

An estimated eleven days will be allocated for the full construction and development process:

- 3 days for physical assembly
- 4 days for coding and system integration
- 4 days for testing and adjustments

Data Analysis

The data will be measured in Voltages for 100 seconds for each gas/environment tested. The first 30 seconds no gas is intorduced to the sensors. Between 30 and 40 seconds the gas/change in environment will be introduced. Then after the 40 second mark the gas introduced will be taken away or the mechanism releasing it will be turned off. From the 40 second mark to 100 seconds the values will be read with the sensors still in place. When normal air flow is tested nothing will be changed in the gas sensors envirnment. The analog pins of the gas sensors will send voltages out based on the amount gas particles they detect. Thus if they send 0.1V, it means almost no harmful gasses are detected if any. If they send above 1V it means they detect a harmful amount of the dangerous gasses they can detect. (CO, Butane, Propane, LPG) The microcontroller (Esp32) can only read voltages from 0V to 3.3V. Thus two voltage dividers will be used to downstep the voltage that is sent from the gas sensors. A voltage divider for each sensor. Will use a 6.8 kΩ and 10 kΩ resistor for each voltage divider. The voltage output between the resistors will be a lower voltage, which the microcontroller will be able to read. The voltage read is then converted/adjusted back to its original value sent by the analog output pin in the code. The values are further explained in the diagrams' descriptions below.

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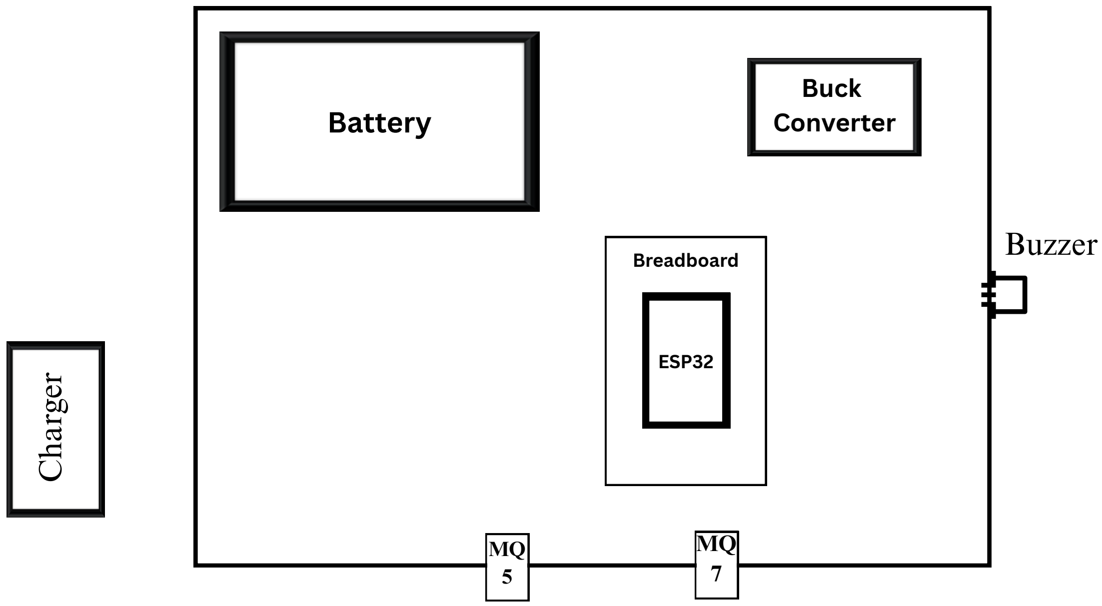
The following values will be given and tested for by the gas detector:

- Natural Air
  - The model wil be put in an area with normal air flow. This is used to compare with other values.
- Lighter
  - Releases Butane gas. Will releasse only the gas and not light it 5cm away from the sensors.
- Candle Flame Smoke
  - Will measure the smoke of the candle smoke when the fire is put out. Amounts of CO is expected to be detected. The smoke will flow through the sensors. The candle itself will be ±10cm and the smoke will be flowing through the sensors.
- Gas Stove
  - The sensors will be put 15cm above the gas stove's plate. Then
  - Releases LPG
- The gas released by an LPG gas pipe itself
  - Contains a variety of the gasses to be detected.
- Charcoal or Wood Embers: Safely ignite a small charcoal briquette or piece of wood outdoors. Once it's glowing (not roaring flame), position the sensor 10–20 cm away —charcoal emits significant CO.

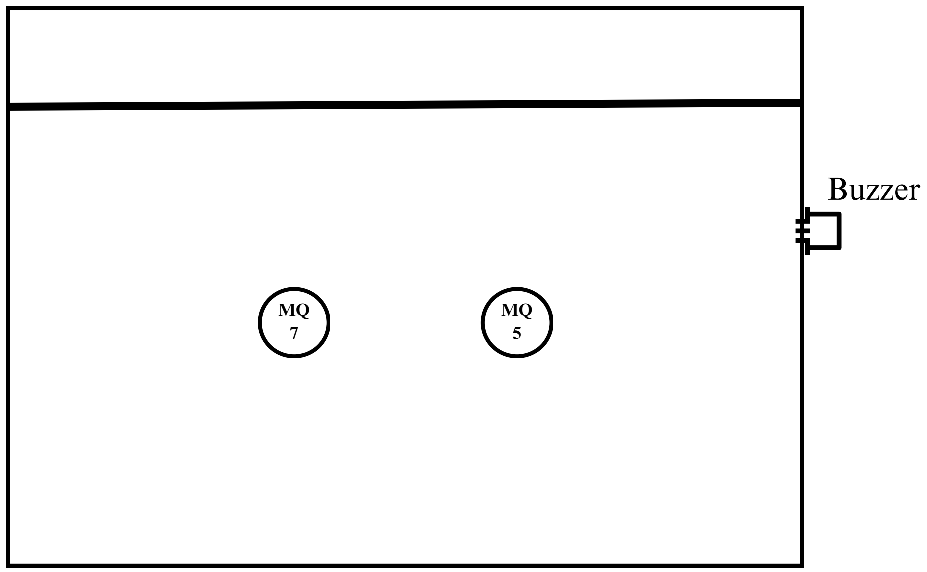
Preliminary Designs

Draft designs and wiring diagrams will be created and added to this section to visualize the layout of components and enclosure.

Physical Prototype 1:

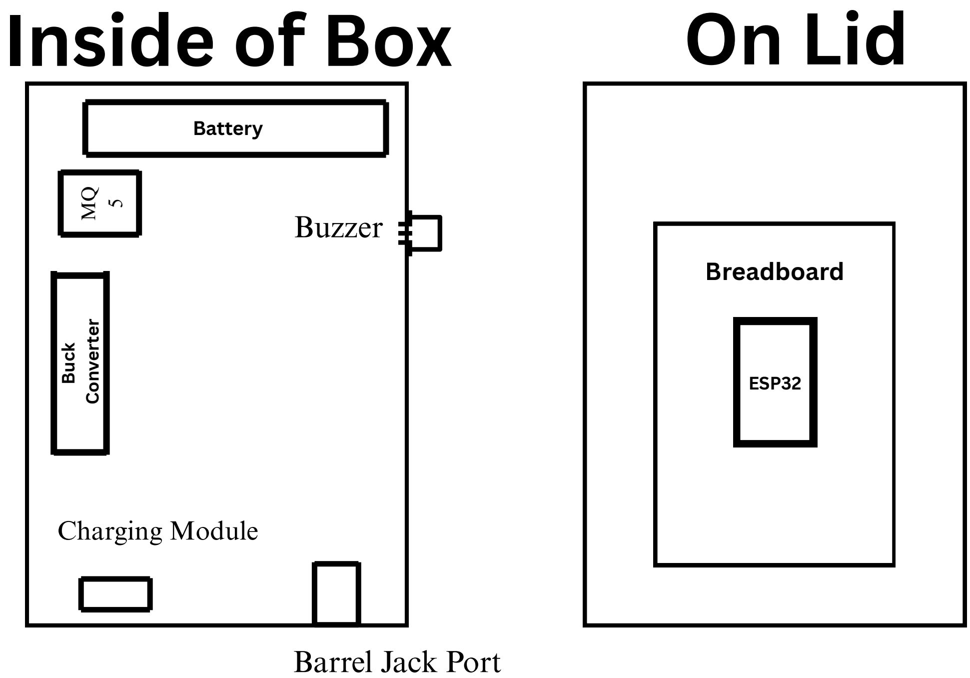


Inside First Physical Prototype

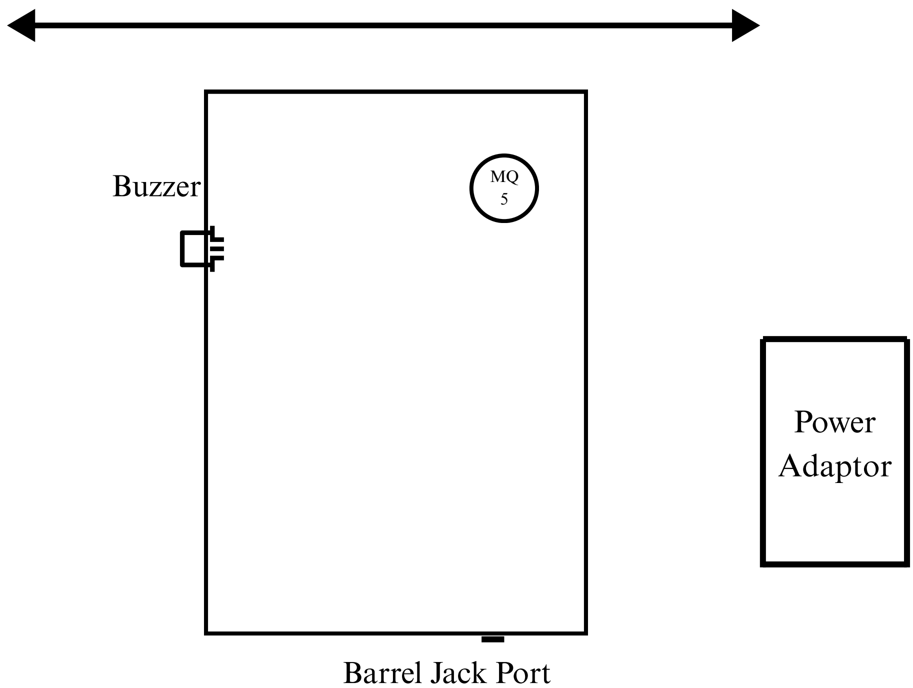


Front First Physical Prototype

Physical Prototype 2:



Inside Final Physical Prototype



Front Final Physical Prototype

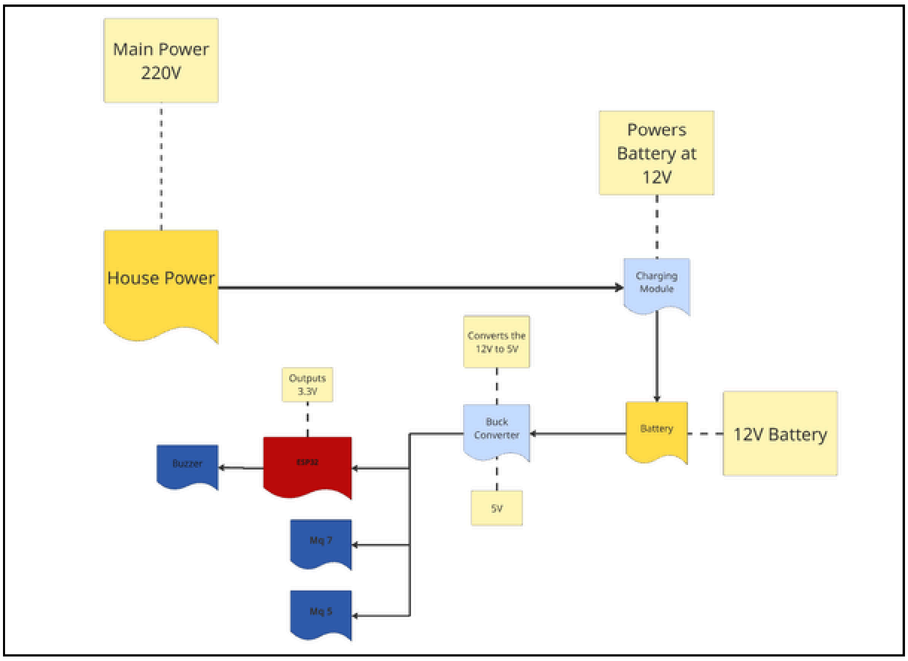
Components Sizes:
Physical Prototype 1

Component	Length (mm)	Width (mm)	Height (mm)
Battery	113	70	85
Charger	197	93	49
ESP32	50	25	14
Buck Converter	65	47	10
MQ 5	32	20	22
MQ 7	32	20	22
Buzzer	33	13	9
Breadboard	83	55	9

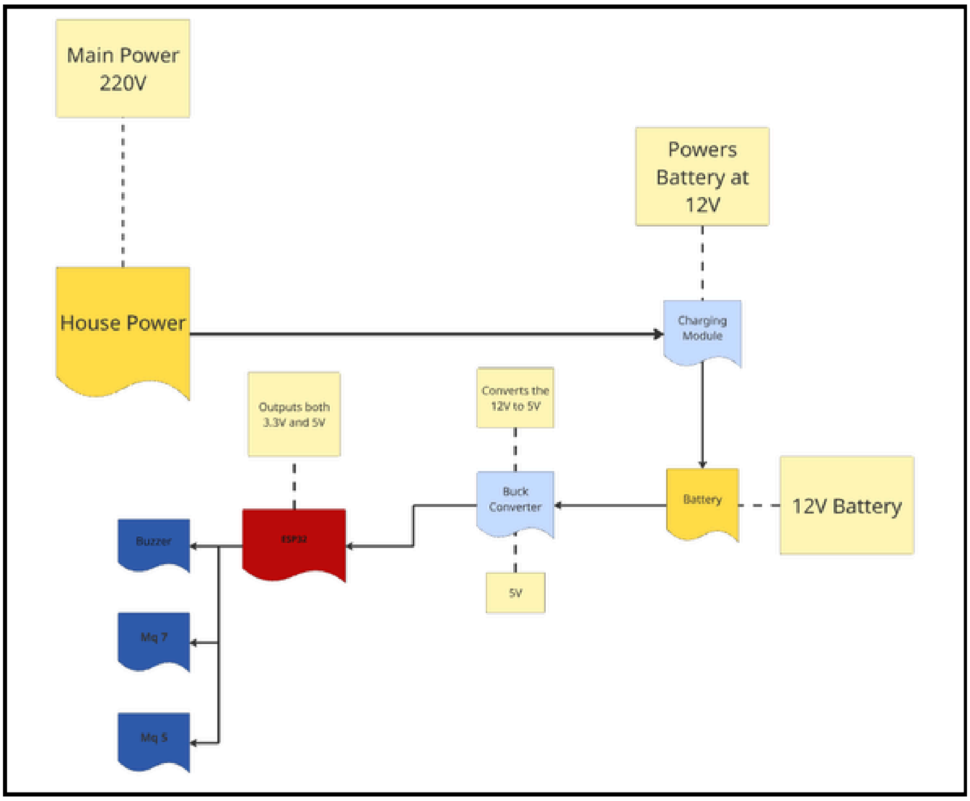
Physical Prototype 2

Component	Length (mm)	Width (mm)	Height (mm)
Casing	115	75	52
Battery	51.5	34	10
Charging Module	25	17.3	5
Power Adaptor	70	50	30
ESP32	50	25	14
Buck Converter	65	47	10
MQ 5	32	20	22
Buzzer	33	13	9
Breadboard	83	55	9
Barrwl Jack Port	14.8	13	11

Phase 1 Power:

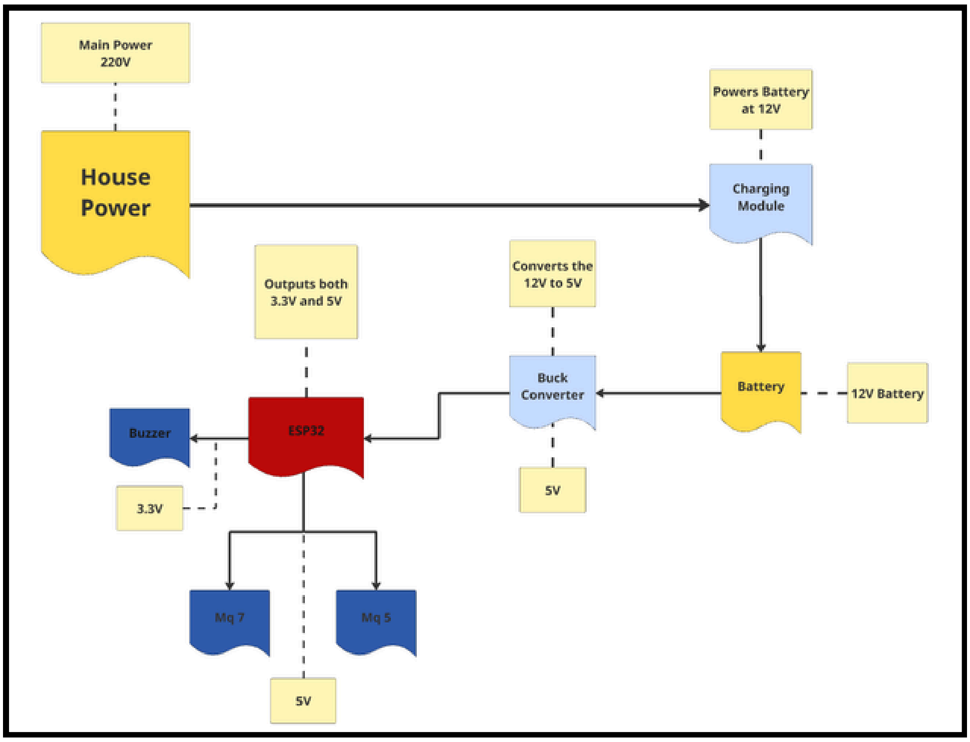


Phase 2 Power:



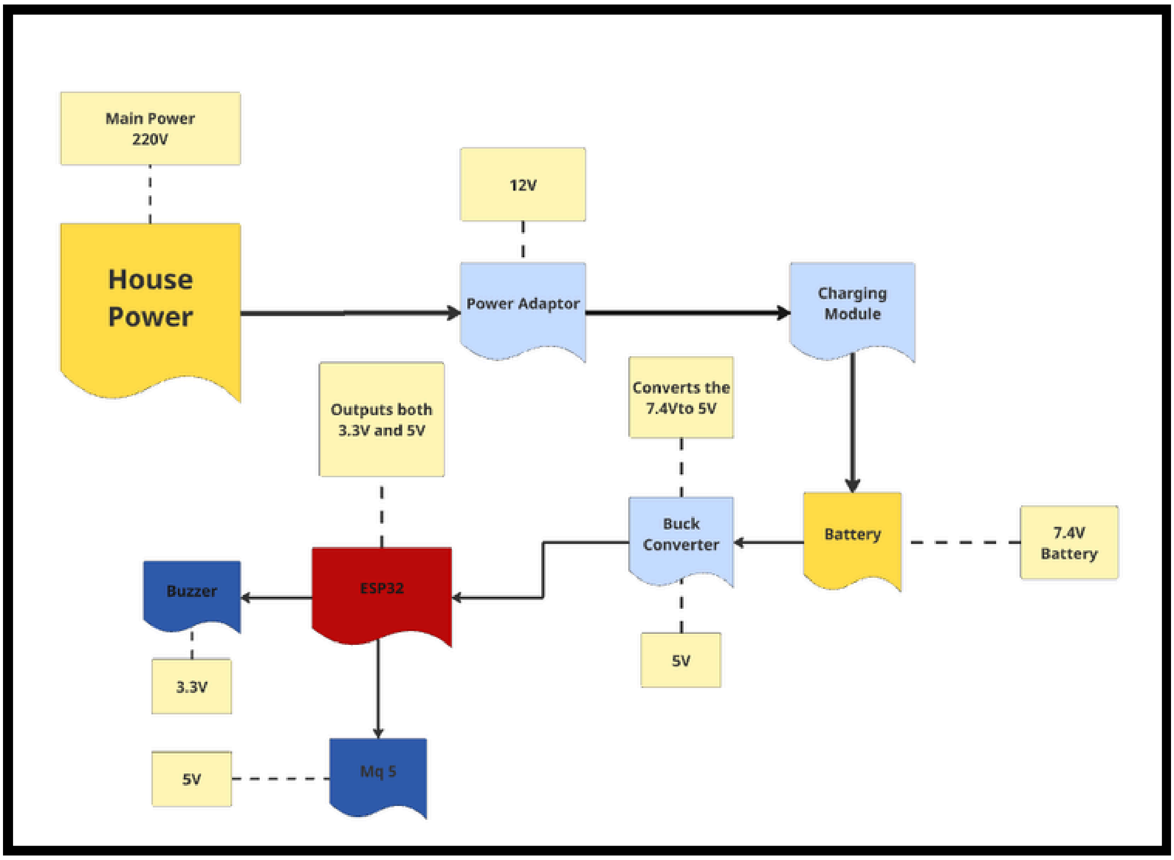
Physical Prototype 1

Phase 3 Power:

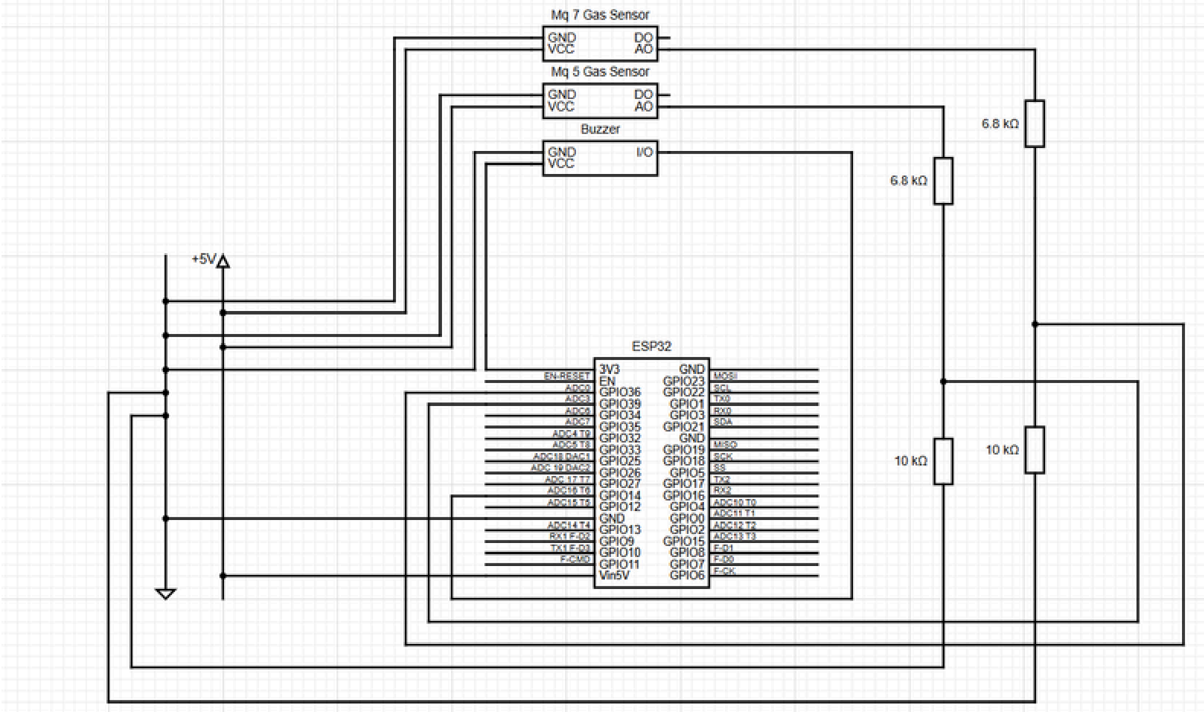


Final (Physical Prototype 2)

Phase 4 Power:



Wiring Diagram:



Explanation Of Diagrams

Power Diagrams:

These diagrams show how the powering of the components work. They show how the voltage will be handled across all components.

Wiring Diagram:

This diagram was made for the first prototype, and it mostly still relevant for the final prototype as the only difference is that the MQ7 gas detector and its voltage divider is no longer present in the system.

This is the main diagram that shows how the wiring/connections between the microcontroller and components look. It also shows the voltage divider used. (Mentioned above) The wiring diagram stays consistent between the two prototypes.

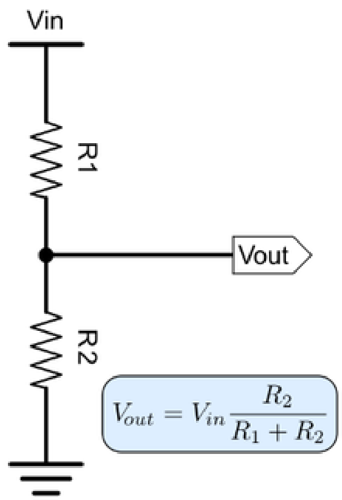
The following will describe the functions of the different pins/components:

- 5V Rail
  - The rail that gives the 5V positive for components.
  - Connects with the Vin5V for power supply.
  - Supplies VCC pins.
- GND Rail
  - Gives ground/negative for the whole circuit.
  - 0V
- AO
  - Sends electrical signal to GPIO pins so the microcontroller can read them and convert the reading to voltage.
  - Can output 0V-5V.

- VCC
  - Positive Terminal
- GND
  - Negative Terminal
- GPIO
  - Can act as an input pin or output pin for the microcontroller.
  - Can read from 0V to 3.3V
- Vin5V
  - The microcontroller can be supplied with 5V via this pin. It is supplied already via its USB c 2.0 port, so now it can supply 5V via this pin.
- 3V3
  - Outputs 3.3V.

Voltage Divider:

The analog pins can output 0V-5V. The microcontroller can only read 0V-3.3V. Thus a voltage divider was used to lower the voltage the Esp32 reads. Esp32 will then convert it back to read the original value via using the following formula:



Vin - 5V maximum

R1 - 6.8kΩ

R2 - 10kΩ

10kΩ and 6.8kΩ will be used. The Vout maximum will then be 2.976190V which is low enough for the Esp32 to read. In the code the voltage will then be adjust by using the voltage ratio between the the maximum of the Vout and Vin5V.

5V/2.976190V = <u>1.68</u>

This value is then used to multiply the readings with to make them accurate to the analog output.

Since carbon monoxide, LPG, and other gasses (usually lpg, propane, butane) are colorless and difficult for humans to detect, these precautions are vital for ensuring the safety of everyone in or near the testing area. The backup gas detector will also serve as an added safeguard in the event that gas escapes unexpectedly or is carried by a gust of wind.

Ethics

The experiments conducted during the testing of the gas detector will make use of flammable gasses. Safety measures were taken to ensure no harm to researchers.

Safety

Harmful gases will be needed to determine whether the gas detector is functioning properly. With this comes certain risks and safety precautions that must be addressed before testing. These risks include:

- **Gas poisoning**
- **The gas leak getting out of control**
- **A possible explosion caused by a spark igniting the gas**
- **Accidental inhalation of the gas**

These risks can be minimized by using specific precautions and testing methods:

- **Only a very small amount of gas will be released at a time to reduce the risk of poisoning**
- **Safety equipment will be on standby to quickly seal the leak, such as a rubber cork to plug the pipe or gas outlet**
- **Tests will be conducted in a controlled and safe environment where no ignition sources are present (e.g., accidental sparks)**
- **Proper gas masks will be worn during testing, with additional masks available as backups**
- **Testing will take place in a well-ventilated area or outdoors**

The following safety measures will be of utmost importance during testing:

- **Wearing appropriate personal protective equipment (PPE), such as goggles, gloves, gas masks, and clothing that covers the arms, torso, and legs**
- **Having safety gear on hand in case of emergencies, including fire extinguishers, first-aid kits, and additional gas masks**
- **Ensuring constant parental supervision throughout the entire testing phase**
- **Keeping a second gas detector nearby to detect any unnoticed leaks**

Time Frames

Term 2

Presentation of project idea to teachers for approval.

25–26 May

Research will commence to identify the components that will be most suitable for the gas detector.

26 May–22 June

Testing will take place on the coding using C++. Experiments will be carried out to determine how the code will be integrated with the hardware of the gas detector. The introduction section of the report will also be drafted during this period.

22–24 June

Further research will be conducted on C++, and the appropriate microcontroller will be chosen for our needs. Revisions and refinements will be made to the introduction based on this updated direction.

2–3 July

Necessary parts will be ordered, and the formatting of the remaining project documentation will be completed in preparation for the following stages of the research plan and final report.

13 July

All available parts will be brought together to commence the building of the gas detector.

14 July

Rest of the parts are expected to arrive. Building of the gas detector will end.

15–17 July

All coding will be implemented in the gas detector and, if necessary, further calibration and adjustments will be made to the gas detector. Diagrams will be made of the electrical work.

17–19 July

Testing of the gas detector will commence. It will be tested on whether it actually works, what gasses it struggles to detect, and the amount of gas that has to be released in order for the alarm to go off.

20-21 July

Adjustments and overview of the report file, research plan, and gas detector itself will happen. Spelling and grammar will be checked, information will be checked to ensure it is still relevant and data will also be checked to ensure it is still accurate.

21-24 July

Final changes will be made to the research plan and report file. Finalisation of all documentation will happen.

25-28 July

Complete poster notes and pages. Prepare to show educators the project.

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